

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 16: The "Sliding Window" Analogy (Manual Harris Corner Detector)

ACTIVITY CONCEPT & OVERVIEW

Concept: Corner detection in image processing (specifically the Harris Corner Detector) functions by evaluating how much local pixel intensities change when shifting a small patch/window (u, v) over an image region.

OBSERVATION & ANALYSIS

Region Type	Shift Movement Direction	Observed Intensity Change	Mathematical Response
Flat Region	Shifting in any direction (u, v)	No intensity change (remains uniform white or black).	Small change in all directions ($E(u,v) \approx 0$).
Edge Region	Shifting along the edge line	No intensity change parallel to edge; strong change perpendicular to edge.	Large change in one direction only.
Corner Region	Shifting in any direction	Significant intensity variation observed regardless of shift direction.	Large change in all directions ($E(u,v)$ is large).

WORKSHEET POST-ACTIVITY QUESTIONS & SOLUTIONS

1. 1. How does the sliding window physical analogy explain the Harris Corner Detection response function?

The sliding window represents the local patch W . The Harris Corner Detector mathematically measures the variation $E(u,v) = \sum W(x,y) [I(x+u, y+v) - I(x,y)]^2$. A corner is detected when shifting the window in *all* directions yields a large change in intensity, corresponding to both eigenvalues (λ_1, λ_2) of the auto-correlation matrix being large.

2. 2. Why does shifting a window along an edge produce zero intensity change in one direction?

Along the edge line, pixel values remain constant (homogeneous). Movement parallel to the edge orientation introduces no gradient variation, which illustrates the *aperture problem*.